

SUMIT PAUL

Bangalore, India

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SUMMARY

Software engineer with hands-on expertise in full-stack development and AI-powered application architecture. Proficient in building scalable microservices, backend REST APIs, and modern frontend solutions using React, Node.js, FastAPI, and cloud technologies. Strong problem-solver experienced in real-time data processing, LLM integration, secure authentication, and production-grade system design.

EXPERIENCE

Hackathon Participant – Smart India Hackathon 2025

2025

Bangalore

- Designed system architecture and developed the Flask backend for AyurSutra --- an AI-powered Ayurvedic patient management platform featuring automated therapy scheduling, JWT authentication, EHR integration, and n8n workflow automation using React, Python Flask, and MSSQL. Qualified for top **15** teams in Grand Finale.
- Applied object-oriented design and agile development practices in a multi-tiered distributed system under competitive time constraints alongside a cross-functional team.

PROJECTS

Community Hero – AI-Powered Civic Engagement Platform

Jun 2026- Jul 2026

(React • TypeScript • Firebase • Cloud Functions • Firestore • Gemini AI • Leaflet • Tailwind CSS)

- Architected full-stack civic reporting platform supporting **1,000+** concurrent users with Gemini AI for automatic issue categorization, production-grade Firestore schema with fine-grained security rules, role-based access control (RBAC) spanning **3+** role levels, and geo-tagged reporting via Leaflet interactive mapping.
- Built real-time analytics dashboard tracking **500+** reported issues, community verification workflow, hotspot prediction engine using Geohashing (nightly cron job), and citizen reputation/gamification system enabling hyperlocal infrastructure issue tracking with geographic precision.

FinPilot AI – Smart Expense Tracker

Sept 2025 - Dec 2025

(SvelteKit • Node.js/Express • FastAPI • MongoDB • Groq LLM • OCR.Space • Docker • Redis • Tailwind CSS)

- Built production-grade platform processing **50+** receipt formats with receipt OCR scanner (OCR.Space + Groq Llama 3.3-70b LLM, **95%+** accuracy), smart bank SMS bulk importer parsing **15+** Indian bank formats with regex-based extraction (supporting INR, USD, EUR, GBP), and AI financial advisor with offline fallback; designed **3-tier** microservices architecture with SvelteKit frontend, Express REST API (**15+** endpoints), FastAPI microservice, and JWT authentication.
- Implemented Chart.js analytics dashboards (monthly trends, category breakdown, weekly comparisons) with 3D visualizations, multi-currency support (**4** currencies: INR, USD, EUR, GBP), containerized deployment via Docker Compose (MongoDB + Redis), and real-time financial health scoring (**0-100** scale).

Image Steganography

Dec 2024- Feb 2025

(Java, Swing, LSB Steganography, AES-256/CBC Encryption, SHA-256 Hashing)

- Engineered a 1,277-line Java Swing application implementing LSB steganography with AES-256/CBC encryption and SHA-256 password hashing, supporting text and audio concealment across JPG, PNG, and BMP formats — built without third-party libraries.
- Achieved **99.2%** embedding capacity with zero perceptual quality loss across **3** image formats; validated security against LSB steganalysis attacks using statistical analysis.

Weather Forecast Website

Mar 2024 - Apr 2024

(HTML5, CSS3, Vanilla JavaScript, OpenWeatherMap API, Geolocation API)

- Built a responsive dark-themed weather app with glassmorphism UI (**12px** backdrop blur) featuring automatic HTML5 Geolocation detection and global city search; integrated dual OpenWeatherMap API endpoints for current weather plus **40+** weather parameters (temperature, humidity, pressure, wind speed/direction), hourly 3-hour forecast, and extended 4-day forecast with dynamic DOM generation.
- Engineered mobile-first CSS Grid/Flexbox layout with **3** responsive breakpoints (mobile $\leq 480\text{px}$, tablet $\leq 768\text{px}$, laptop $\leq 1024\text{px}$); implemented **5+** micro-interactions, keyframe animations, and Google Fonts (Roboto + Orbitron); optimized for **<2s** load time with CSS3 variables and custom properties.

TECHNICAL SKILLS

Programming Languages:	Java, Python, C, JavaScript, TypeScript
Frontend Technologies:	React, SvelteKit, Tailwind CSS, HTML5, CSS3, Responsive Design
Backend Technologies:	Node.js, Express.js, FastAPI, REST APIs, JWT Authentication
Databases & Storage:	MongoDB, MySQL, SQLite, NoSQL
Developer Tools:	Git, GitHub, Docker, VS Code, MongoDB Atlas, Figma
Core Computer Science:	Data Structures & Algorithms, OOP, System Design, API Design
Data Visualization:	Tableau, Power BI, Matplotlib, Microsoft Excel.

EDUCATION

New Horizon College of Engineering

2023–2027

B.E – Computer Science and Engineering (VTU) – CGPA – 9.59

Bangalore, Karnataka

CERTIFICATIONS & ACHIEVEMENTS

- Data Structures and Algorithms with Java (Alpha) – Apna College
- Introduction to Networking and Cloud Computing – Microsoft (2024)
- Oracle University – Oracle Certified Foundations Associate and Generative AI Professional
- Vibe2Ship – National-Level AI Hackathon Participant | Coding Ninjas × Google for Developers (2026).

COURSEWORK AND INTERESTS

Relevant Coursework: Data Structures & Algorithms — Object-Oriented Programming — DBMS — Computer Networks — Cyber Security — Data Mining & Machine Learning — Web Development

Interests: Sketching and Drawing, Dancing, Learning New Programming Languages

Languages: Hindi (Fluent), Bengali (Bilingual), Assamese (Bilingual), English (Professional Proficiency)